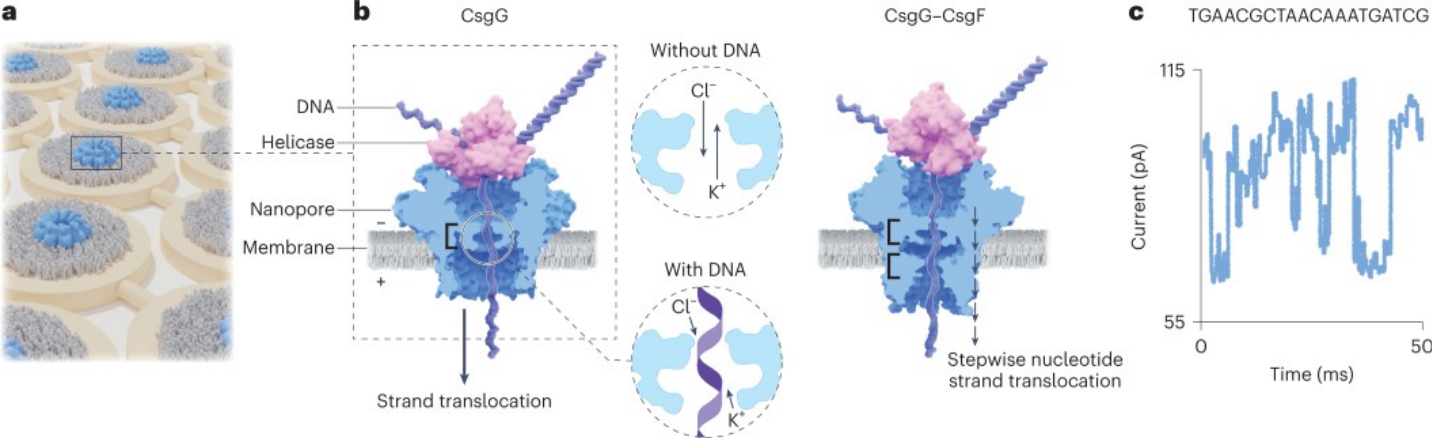

a
b
CsgG
DNA
Helicase
Nanopore
Membrane
Strand translocation
Without DNA
Cl⁻
K⁺
With DNA
CsgG–CsgF
Stepwise nucleotide strand translocation
c
TGAACGCTAACAAATGATCG
115
55
Current (pA)
0
50
Time (ms)